

YU WING NG, RAYMOND

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

FULL-STACK DEVELOPMENT | AI AGENT ENGINEERING | CLOUD PLATFORMS

EDUCATION

Web Development & Internet Applications Diploma

Jan 2023 - Apr 2024

Algonquin College, Ottawa | Dean's Honours List: 3.95/4.0

Master of Arts in Journalism

Sep 2019 - May 2021

The Chinese University of Hong Kong

Bachelor of Social Science

Sep 2015 - May 2018

City University of Hong Kong

PROGRAM-RELATED SKILLS

AI SDKs & Model Integration

Hands-on integration of **OpenAI**, **Anthropic (Claude)** and **Google Gen AI (Gemini)** SDKs in JavaScript/TypeScript applications; tool/function calling, streamed responses, multimodal inputs and structured JSON outputs.

Provider-aware request configuration, context handling and model selection based on task complexity, latency and inference cost.

Agent Orchestration & Reliability

Agent workflow design, multi-step tool execution, context/state management and schema validation. Reliability patterns covering bounded retries with exponential backoff, timeout handling, rate-limit recovery, controlled fallbacks and idempotent operations; separation of deterministic business logic from model-generated decisions.

Caching Policy & Inference Efficiency

Provider-specific prompt/context caching: **OpenAI prompt-prefix reuse**, **Claude cache-control breakpoints** and **Gemini implicit/explicit context caching**. Stable prompt/tool-schema design, cache lifetime and invalidation behaviour, cache-hit and token-usage monitoring, and cache-read/write/storage cost trade-offs; distinction between prompt caching and application response caching.

RAG, MCP & Trustworthy AI

Retrieval-augmented generation (RAG), Model Context Protocol (MCP), structured knowledge retrieval and metadata filtering. Evidence-linked outputs, data-sufficiency gates, permission-aware tool access, human approval and server-enforced usage controls for AI-assisted education workflows.

Cloudflare & Edge Architecture

Cloudflare Workers, **Hono**, **Durable Objects**, **Workflows**, **AI Search and Tunnel**; serverless API orchestration, persistent workflow state and background-task design. Architectural separation of lightweight edge requests from CPU-intensive, containerised document processing.

Full-Stack Engineering & Data

TypeScript/Javascript, React, Node.js/Express, RESTful APIs and MongoDB; nested JSON modelling, authentication, role-based access and institutional workflows. Supporting experience with MySQL, Python/Django and C#/.NET.

Containers & Multi-Cloud Hosting

Docker, **Docker Compose** and Linux-based hosting; service isolation, environment configuration and deployment troubleshooting. Working knowledge of **Google Cloud Platform (GCP)**, **Microsoft Azure** hosting and Cloudflare Containers deployment considerations.

CI/CD, Testing & Observability

Git/GitHub and **GitHub Actions CI/CD**; automated build, test and deployment workflows, environment/secrets management, API testing and end-to-end testing with Playwright, Mocha and Chai. Logging, error diagnosis and AI usage monitoring.

ENTERPRISE PROJECT EXPERIENCE

ExamPedita | AI-Powered Education Platform

An all-in-one AI-powered education ecosystem connecting students, educators and institutions—from content creation and delivery to assessment, learning support and resource monetisation.

FUNDING RECOGNITION

Received funding support through the PolyU Micro Fund and the HKSTP Ideation Programme.

- **Full-stack ownership:** Develop the React/TypeScript frontend and Node.js/Express backend, with MongoDB data models and API integrations for educators, students and institutional administrators.
- **Multi-provider AI:** Integrate OpenAI, Claude and Gemini SDKs for question analysis, assessment support and contextual explanations, accounting for provider-specific context, caching and inference-cost behaviour.
- **Structured retrieval & MCP:** Connect structured HKDSE resources and institutional knowledge to retrieval workflows; expose reusable tools through MCP so AI clients can query education resources.
- **Institutional capabilities:** Support class rosters, role/permission controls, approval workflows, reporting and Google Classroom integration.
- **Cloud architecture:** Develop a hybrid architecture combining Cloudflare edge services with Docker-based processing, separating interactive requests from asynchronous AI work and resource-intensive document handling.
- **Responsible agent design:** Define evidence-linked analysis, deterministic data gates, human approval and cost controls for learning-trajectory and intervention workflows.

WORK EXPERIENCE

Founder | ExamPedita Limited

May 2024 - Present

- Lead product strategy and hands-on development, translating education needs into the technical roadmap and managing implementation, releases and ongoing improvements.
- Drive marketing, product positioning, customer outreach and business development; manage day-to-day operations and coordinate business priorities with engineering delivery.

Executive Officer | Tung Chung Catholic School

Aug 2019 - Jul 2022

- Coordinated school administration across financial and personnel matters, procurement and tendering, and premises maintenance, working with internal teams and contractors to deliver school improvement projects.
- Led 3 school office administrative staff and 7 school support workers, managing workloads, preparing duty rosters and coordinating day-to-day administrative and operational work.

REFERENCES

Michael Alan Eisenbraun Professor, ICT, Applications & Programming Algonquin College, Ottawa, ON eisenbm@algonquincollege.com	Hao Yan Chau Vice Principal Tung Chung Catholic School, Hong Kong chy@tccs.edu.hk
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ADDITIONAL SKILLS

Strengths: Self-directed learning, analytical problem-solving, independent delivery and cross-functional teamwork.

Communication: Professional English communication and effective collaboration with technical and non-technical stakeholders.